Problem 4

Caleb Hessing

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$$F_n = \begin{cases} 1 & \text{if } n = 1 \\ 1 & \text{if } n = 2 \\ F_{n-1} + F_{n-2} & \text{if } n \geq 3 \end{cases}$$

Proposition 1.

$$\sum_{j=1}^{m} F_{2j-1} = F_{2m}$$

Proof. The proof is by induction on m.

BASE CASE: Consider the case for $m = 1$. Observe,

$$\begin{aligned} \sum_{j=1}^{1} F_{2j-1} &= F_{2(1)-1} \\ &= F_{2-1} \\ &= F_1 \\ &= 1 \end{aligned}$$

and,

$$F_{2(1)} = F_2 = 1$$

Thus, the result holds for the base case $m = 1$.

INDUCTIVE CASE: Suppose the result holds for some k. That is,

$$\sum_{j=1}^{k} F_{2j-1} = F_{2k}$$

□

Proposition 2.

$$\sum_{j=1}^{m} F_{2j} = F_{2m+1} - 1$$

Proof. The proof is by induction on m.

An impressive proof.

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